

An Excursion into Toeplitz Quantization on Symmetric Domains

Presenta:

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- 1 The Quantization Problem
- 2 Reproducing Kernel Hilbert Spaces and Toeplitz Operators
- 3 An Example in $\mathbb{C}^n$

The Quantization Problem

Quantization is generally understood as the transition from classical mechanics to quantum mechanics. Starting from a classical system, one often seeks to formulate a quantum theory that, in a suitable limit, reduces back to the original classical system. Mathematics has provided various tools to tackle this challenge.

- **Classical Physics:** Observables are commuting functions.
- **Quantum Mechanics:** Observables are operators on a Hilbert space that do **not** commute.

Canonical Quantization

The original concept of quantization (nowadays generally known as *canonical quantization*) dates back to the times of Weyl, von Neumann, and Dirac.



Figure: Hermann Weyl

It consists of attempting to assign to the observables of classical mechanics, which are functions on $\mathbb{R}^{2n}$ (the phase space), self-adjoint operators Q_f on the Hilbert space $L^2(\mathbb{R}^n)$ such that:

Quantization Axioms

- (A1) The mapping $f \mapsto Q_f$ is linear;
- (A2) (*Von Neumann's rule.*) For any polynomial $\phi : \mathbb{R} \rightarrow \mathbb{R}$, we must have

$$Q_{\phi \circ f} = \phi(Q_f).$$

(In particular, $Q_1 = I$.)

- (A3) $[Q_f, Q_g] = -\frac{i\hbar}{2\pi} Q_{\{f,g\}}$, where

$$\{f, g\} = \sum_{j=1}^n \left(\frac{\partial f}{\partial p_j} \frac{\partial g}{\partial q_j} - \frac{\partial f}{\partial q_j} \frac{\partial g}{\partial p_j} \right)$$

is the Poisson bracket of f and g .

IT IS IMPOSSIBLE

Since it is not possible to have a perfect quantization, the mathematical community has developed various approaches:

- **Geometric Quantization**

- Associates Hilbert spaces to symplectic manifolds via pre-quantization and polarization.

- **Deformation Quantization**

- Deforms the algebra of classical observables using a noncommutative product (the star product $\star$), without constructing explicit Hilbert spaces.

- **Berezin-Toeplitz Quantization (Our approach)**

- Employs pure functional analysis on reproducing kernel Hilbert spaces (RKHS) to construct a deformation quantization.

Relaxing the commutator axiom to:

$$[Q_f, Q_g] = -\frac{i\hbar}{2\pi} Q_{\{f,g\}} + O(\hbar^2).$$

The search for a concrete Hilbert space is abandoned, and it is assumed to be contained in the following space:

the ring of formal power series in the parameter $\hbar$

$$\mathcal{A}_\hbar = C^\infty(M)[[\hbar]] = \left\{ \sum_{k=0}^{\infty} \hbar^k f_k \mid f_k \in C^\infty(M) \right\}$$

Moyal or Star Product

Let $f, g \in C^\infty(M)$

$$f \star_{\hbar} g = \sum_{j=0}^{\infty} \hbar^j C_j(f, g)$$

where the bilinear operators C_j satisfy:

$$\begin{aligned} C_0(f, g) &= fg, \\ C_1(f, g) - C_1(g, f) &= -\frac{i}{2\pi} \{f, g\}, \\ C_j(f, 1) &= C_j(1, f) = 0 \quad \forall j \geq 1. \end{aligned}$$

Note: “Formal” means that $f \star_{\hbar} g$ is not required to converge for a given value of $\hbar$. It is only required that the coefficients $C_j : \text{Obs} \times \text{Obs} \rightarrow \text{Obs}$ are well-defined mappings and satisfy the relations that make $\star_{\hbar}$ formally associative.

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Definición 2.1

We say that a Hilbert space $\mathcal{H} \subseteq \mathcal{F}(X, \mathbb{C})$ is a **Reproducing Kernel Hilbert Space (RKHS)** if for every $x_0 \in X$, the evaluation functional at x_0

$$\begin{aligned} F_{x_0} : \mathcal{H} &\rightarrow \mathbb{C} \\ f &\mapsto f(x_0) \end{aligned}$$

is bounded.

The Riesz representation theorem implies that for every $x \in X$ there exists a unique element $K_x \in \mathcal{H}$ such that

$$f(x) = \langle K_x, f \rangle \text{ for every } f \in \mathcal{H}.$$

Since for every $x \in X$ we have $K_x \in \mathcal{H}$, we similarly obtain that

$$K_x(y) = \langle K_y, K_x \rangle$$

The choice of the RKHS to carry out the quantization (via the Toeplitz calculus) is directly determined by the geometry of the manifold that models the classical phase space.

We will fundamentally consider two spaces:

- **Fock Spaces**
- **Bergman Spaces**

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Weighted Fock Spaces and Toeplitz Operators

Let $\mathbb{C}^n$ be the vector space of all ordered n -tuples of complex numbers; for $\lambda > 0$, we denote by μ_λ the measure

$$d\mu_\lambda(z) = \frac{1}{(\pi\lambda)^n} e^{-\frac{\|z\|^2}{\lambda}} dz$$

where dz denotes the $2n$ -dimensional Lebesgue measure.

Definition

The **Fock space on $\mathbb{C}^n$ with weight λ** is the space $\mathcal{F}_\lambda(\mathbb{C}^n)$ of holomorphic functions on $\mathbb{C}^n$ that are square-integrable with respect to the measure μ_λ , that is

$$\mathcal{F}_\lambda(\mathbb{C}^n) := \left\{ f \in O(\mathbb{C}^n) : \int_{\mathbb{C}^n} |f(z)|^2 d\mu_\lambda(z) < \infty \right\}$$

Fock Kernel:

$$K_\lambda(z, w) = e^{\frac{z \cdot \bar{w}}{\lambda}}, \quad z, w \in \mathbb{C}^n.$$

Fock Projection:

$$\begin{aligned} P_\lambda f(z) &= \int_{\mathbb{C}^n} K_\lambda(z, w) f(w) d\mu_\lambda(w), \quad f \in L^2(\mathbb{C}^n, \mu_\lambda), \quad z \in \mathbb{C}^n \\ &= \int_{\mathbb{C}^n} e^{\frac{z \cdot \bar{w}}{\lambda}} f(w) d\mu_\lambda(w) \end{aligned}$$

Definition

Given $\varphi \in L^\infty(\mathbb{C}^n)$ and $\lambda > n$, the **Toeplitz operator with symbol φ on $\mathcal{F}_\lambda(\mathbb{C}^n)$** is the bounded linear operator

$$T_\varphi^{(\lambda)} : \mathcal{F}_\lambda(\mathbb{C}^n) \longrightarrow \mathcal{F}_\lambda(\mathbb{C}^n)$$

defined as

$$T_\varphi^{(\lambda)} := P_\lambda \circ M_\varphi|_{\mathcal{F}_\lambda(\mathbb{C}^n)}$$

where

$$M_\varphi : L^2(\mathbb{C}^n, \mu_\lambda) \longrightarrow L^2(\mathbb{C}^n, \mu_\lambda)$$

denotes the multiplication operator by φ .

- The mapping $f \mapsto T_f$ is linear.
- $\|T_f\| \leq \|M_f\| = \|f\|_\infty$; in particular, T_f is bounded for $f \in L^\infty$.
- $T_1 = I$, the identity operator on $\mathcal{F}$.
- Toeplitz operators behave well when taking adjoints: $T_f^* = T_{\bar{f}}$.

Taking $\alpha = \frac{1}{\lambda}$ we have the following relations:

$$T_{z_j} = M_{z_j}$$
$$T_{z_j}^* = \frac{1}{\alpha} \partial_j$$

Furthermore, in the case of $\mathbb{C}$:

- $[T_z, T_{\bar{z}}] = \left[z, \frac{\partial}{\partial \bar{z}} \right] = -I,$

Moreover, writing $z = p + iq$ we have that:

$$[T_p, T_q] = \frac{\hbar}{2\pi i} I, \quad \alpha := \frac{\pi}{\hbar}.$$

If f and g are polynomials in z and $\bar{z}$ we have that:

$$[T_f, T_g] = \frac{\hbar}{\pi} T_{\frac{\partial g}{\partial \bar{z}} \frac{\partial f}{\partial \bar{z}} - \frac{\partial f}{\partial \bar{z}} \frac{\partial g}{\partial \bar{z}}} + O(\hbar^2).$$

$$T_f T_g = T_{fg} - \frac{1}{\alpha} T_{(\partial f)(\bar{\partial} g)} + O(\hbar^{-2}).$$

The mapping

$$Q_\lambda : L^\infty(\mathbb{C}^n) \longrightarrow \mathcal{L}(\mathcal{F}_\lambda(\mathbb{C}^n))$$
$$\varphi \longmapsto T_\varphi^{(\lambda)}$$

produces a quantization on $\mathbb{C}^n$. Where the star product is given by:

$$f * g = \sum_{m=0}^{\infty} \hbar^m C_m(f, g),$$
$$C_m(f, g) = \frac{(-1)^m}{\pi^m} \sum_{j \in \mathbb{N}^n, |j|=m} \frac{1}{j!} [(\partial^j f)(\bar{\partial}^j g)]$$

In the case $n = 1$ it reduces to:

$$f * g = \sum_j \frac{(-1)^j \hbar^j}{j! \pi^j} (\bar{\partial}^j g) \partial^j f$$

Thanks :D

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